



**College of Sciences
Department of Mathematics**

Course Syllabus

Course Title Calculus I

**Course Code
1440133**

Spring Semester - 2021/2022

Course Syllabus

Course Title	Calculus I		
Course Code	1440133		
Credit Hours	3		
Pre-requisite(s)	None		
Co-requisite(s)	N/A		
Semester	Spring	Year	2021/2022
Instructors Name	Abdelaiz Soufyane		
Office Location	M7-105		
Tel. No.	Ext.		
Email	asoufyane@sharjah.ac.ae		
Lecture Times	Mon-Wed: 11-12:15		
Office Hours	Mon 12:30-13:30		

Course Description (as in the catalogue):

Functions, domain and range, examples of functions. Limits and continuity. Derivatives by a variety of techniques including explicit differentiation, implicit differentiation, and logarithmic differentiation. Use these derivatives to study the characteristics of curves. Determine derivatives using implicit differentiation and use to study characteristics of a curve, applications of derivatives in optimization, linearization and graphing, the Mean Value Theorem. Integration, the Fundamental Theorem of Calculus, areas, volumes of solids of revolution, arc length of a plane curve.

Course Learning Outcomes:

By the end of successful completion of this course, the student will be able to:

1. Recognize continuous, differentiable, and integrable functions.
2. Find the limits and test for continuity.
3. Use differentiation rules to differentiate algebraic and transcendental functions
4. Use basic techniques of integration to find particular or general antiderivatives.
5. Sketch the graph of a function using asymptotes, monotonicity, and concavity.
6. Demonstrate the connection between areas, volumes and the definite integrals.

7. Use differentiation and integration to solve real world problems such as rate of change, optimization, and area problems.

Alignment of Course Student Learning Outcomes (CLOs) to Program Student Learning Outcomes (PLOs)

PLOs	CLOs
1. Apply a broad range of core mathematical knowledge and techniques, including advanced calculus, linear algebra, geometry, differential equations, probability and statistics.	1. Recognize continuous, differentiable, and integrable functions. 2. Find the limits and test for continuity. 3. Use differentiation rules to differentiate algebraic and transcendental functions 4. Use basic techniques of integration to find particular or general antiderivatives. 5. Sketch the graph of a function using asymptotes, monotonicity, and concavity.
2. Analyze scientific experiments and research methods to continuously build on existing knowledge with modern and innovative ideas.	
3. Use mathematical knowledge and IT skills to design, implement and enhance computer programs.	
4. Work within a team and communicate with a variety of audiences.	
5. Work with professional and ethical responsibility.	
6. Develop mathematical models to solve real-life problems in a contemporary global and societal context.	6. Demonstrate the connection between areas, volumes and the definite integrals. 7. Use differentiation and integration to solve real world problems such as rate of change, optimization, and area problems.

Weekly Distribution of Course Topics/Contents (according to the textbook)

CLOs	Comments (Suggested problems)	Topic	Section	Week
1,2	1.2: 1-6,7-10,27,28. 1.3: 1-31, 43-65.	- The Concept of Limit - Computation of Limits	1.2 1.3	1
1,2	1.4: 1-28,33-44.	Continuity and Its Consequences	1.4	2
1.2	1.5: 1-32,39-48.	Limits Involving Infinity	1.5	3
3	2.1: 1-8,13-22. 2.2: 1-22,32.	- Tangent Lines and Velocity - The Derivative	2.1 2.2	4
3	2.3: 1-48. 2.4: 1-24. 2.5: 1-48.	- Computation of Derivative: The Power Rule - The Product and Quotient Rules - The Chain Rule Derivatives	2.3 2.4 2.5	5
3	2.6: 1-36. 2.7: 1-43	- Derivatives of Trigonometric Functions. - Derivative of Exponential and Logarithmic Functions.	2.6 2.7	6
3	2.8: 1-19, 25-37 2.10: 1-33	- Implicit Differentiation and Inverse Trig. Func. - The Mean Value Theorem	2.8 2.9	7
3	3.1: 1-31, 35-39,51,53 3.2:1-54 3.3: 1-37	- Linear Approximations and Newton's method - Indeterminate Forms and L'Hopital's Rule - Maximum and Minimum Values	3.1 3.2 3.3	8
3	3.4: 1-37 3.5: 1-45	- Increasing and Decreasing Functions - Concavity and The Second Derivative Test	3.4 3.5	9
5,7	3.6: 1-37 3.7: 1-53	- Overview of Curve Sketching - Optimization	3.6 3.7	10
4	4.1: 1-47	Anti-derivatives	4.1	11
4	4.5: 1-42	The Fundamental Theorem of Calculus	4.5	12
4	4.6: 1-39	Integration by Substitution	4.6	13
6	5.1: 1-17 5.2: 17-37	- Area between Curves - Volumes (The methods of Disks and Washers)	5.1 5.2	14
6	5.4: 1-22	Arc Length of a plane curve	5.4	15
Final Exam				16

Students' Assessment:

Students are assessed as follows:

Assessment Tool(s)**	Date	Weight (%)
Assignments	Weeks 2, 5, 8 and 11	5
Quizzes	Weeks 3, 6, 9 and 12	20
Mid semester exam	Week 8	30
Final Exam	Week 16	45
Total		100
Notes: 1. Exact days of the quizzes, assignments, midterm, and the final will be announced later. 2. We will count the best 3 out of 4 quizzes and hence NO MAKEUP for quizzes.		

** You can modify / add other tools relevant to the course.

Course Outcome Assessment Plan

CLOs	Teaching/Learning Method(s)	Assessment Tool(s)	Performance Indicators
Recognize continuous, differentiable, and integrable functions.	- Lectures - Problem-solving - Group discussion	- Exams - Homework assignments	75% of students achieved cutoff value of 70%
Find the limits and test for continuity.	- Lectures - Problem-solving - Group discussion	- Exams - Homework assignments	75% of students achieved cutoff value of 70%
Compute derivatives.	- Lectures - Problem-solving - Group discussion	- Exams - Homework assignments	75% of students achieved cutoff value of 70%
Evaluate integrals.	- Lectures - Problem-solving - Group discussion	- Exams - Homework assignments	75% of students achieved cutoff value of 70%
Sketch the graph of a function using asymptotes, monotonicity, and concavity.	- Lectures - Problem-solving - Group discussion	- Exams - Homework assignments	75% of students achieved cutoff value of 70%
Calculate areas and volumes of revolution.	- Lectures - Problem-solving - Group discussion	- Exams - Homework assignments	75% of students achieved cutoff value of 70%
Formulate and solve time-related and optimization problems.	- Lectures - Problem-solving - Group discussion	- Exams - Homework assignments	75% of students achieved cutoff value of 70%

Teaching and Learning Resources:

Text Book(s):

Smith, R. T., Minton, R. B. and Rafhi, Z. (2018). Calculus Early transcendental functions. 5th Edition. McGraw Hill, ISBN-13: 978-0077166472

Recommended Readings:

Weir, M. D., Hass, J. and Giordano, F. R. (2017). Thomas' Calculus. 14th Edition. Addison-Wesley. ISBN-13: 978-0134438986

Other Resources:

Reading, library resources, internet (optional), MS-Team Tutorials

Attendance policy:

Attendance is compulsory. A student missing 10% (6 hours) of the total allocated course hours will receive 1st warning notice and a student missing 15% (9 hours) will receive 2nd warning notice. A student missing 20% (12 hours or more) will be forced to withdraw (in accordance with the university regulations).

Plagiarism/Cheating:

Students are expected to do their own work. You are allowed to work on assignments in teams only if specified by the instructor. In other words, students are encouraged to communicate about general principles of the course, but all assigned homework must be done on an individual basis. The instructor is available to provide any assistance that you may need. Cheating is considered a serious offense by the university. You should be aware of the severe penalty for cheating (refer to the student code of conduct published in the university catalogue).

Notes:

Examination Policy & Appeals Against Academic Assessment:

Midterm and Final exams will be administered at the scheduled time as predetermined by the College of Science. No test scores will be dropped, and no re-takes will be offered for poor performance. Absences on the test days must be approved by the course instructor prior to the scheduled test. An unexcused absence from a test may result in a grade of "0" for that test. All initial contact for appeals against academic assessment should be via the course coordinator. Students will be required an appointment to discuss the appeal. If required, the appeal will be escalated to the Head of the Department of the course, and other bodies as per the University of Sharjah standard operating procedure.